SFR-Net: Scattering Feature Relation Network for Aircraft Detection in Complex SAR Images

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***Abstract*— Aircraft detection in synthetic aperture radar (SAR) images plays a significant role in dynamic monitoring and national security. Previous methods have difficulty in obtaining the desirable detection performance due to the interference of complex scenes and diversity of aircraft sizes. In order to solve these problems, we propose an innovative scattering feature relation network (SFR-Net) in this article. First, considering that the strong scattering points of the aircraft in SAR images are usually discrete, we leverage the proposed scattering point relation module to fulfill the analysis and correlation of scattering points. By enhancing the characteristics and relationships among the scattering points, this method is beneficial to guarantee the completeness of aircraft detection results. Second, we design a salient fusion module to adaptively aggregate the features from different layers of SFR-Net with rich semantic information and plentiful details, which can highlight the significant objects with different sizes and enhance the distinguishable features. Third, to reduce the false alarm and improve the localization accuracy, the contextual feature attention is presented to capture the global spatial and semantic information with a large receptive field. Overall, the SFR-Net is designed based on the SAR imaging mechanism and the scattering characteristics of aircrafts. The extensive experiments are conducted on the SAR aircraft detection dataset (AIRD) from the Gaofen-3 satellite to demonstrate the effectiveness of the SFR-Net and also illustrate that our method achieves state-of-the-art performance.**

***Index Terms*— Aircraft detection, contextual feature attention (CFA), feature fusion, scattering point, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) can provide high-resolution images based on the electromagnetic scattering imaging mechanism, and this sensor can operate in all-day and all-weather conditions. With the advancement of airborne SAR and spaceborne SAR imaging technologies, more high-quality SAR images have been obtained. SAR plays a crucial role in various fields [1]–[6], such as terrain detection, target detection, and environment monitoring. In recent years, lots of studies [7]–[11] have made progress in target detection. Among different targets, aircraft detection has essential applications in global regional surveillance and security.

Traditional SAR target detection algorithms mainly include three categories: statistical analysis [12], [13], geometric and structural features extraction [14], [15], and wavelet transformation [16], [17]. At present, the constant false alarm rate (CFAR) [18] is a typical method of statistical analysis and has wide applications. It conducts statistical modeling of background clutter and detects anomalies that do not conform to the clutter distribution. Ai *et al.* [13] proposed a dual-parameter CFAR method, which can remove the outliers based on adaptively truncated clutter statistics in the background window. In order to describe the amplitude of SAR images more accurately, Gierull and Sikaneta [19] developed a statistical model of sea clutter to handle the limited thermal noise, which improves the calculation simplicity. Considering the collinearity and symmetry of the target structure, Gao *et al.* [15] combined the Hough transform to extract the skeleton of targets for further detection. Based on the spatial and frequency characteristics of the SAR images, Tello *et al.* [17] used wavelet transform to perform multiscale target detection. However, in the face of complex background scattering, traditional methods are not very effective due to the poor robustness and the limitation of feature description capabilities.

Recently, with the development of deep learning, a convolutional neural network (CNN) presents a strong feature representation ability in the field of computer vision. The methods based on deep CNNs are introduced to the SAR field [20], [21], and the results demonstrate the excellent performance of deep learning. In terms of target detection, some stepwise algorithms [22], [23] leverage CNN to extract features from SAR images to identify objects. For aircraft targets, the location of the airport is first determined in the SAR images according to thresholds determined by the OTSU algorithm [24], and then, aircraft detection is performed on the area within the airport. Although this kind of method can reduce false alarms to a certain extent, its process does not end to end and is relatively cumbersome.

At the current stage, lots of end-to-end methods based on the anchor-based and anchor-free detectors are widely

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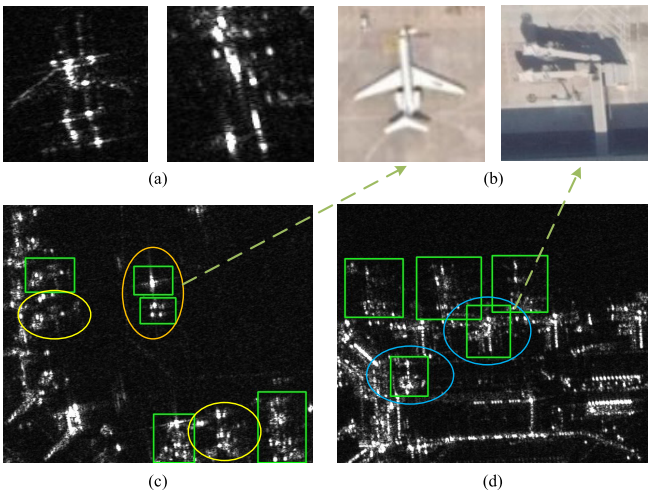


Fig. 1. Detection results of the common anchor-free method. The green boxes, yellow circles, orange circles, and blue circles represent the detection objects, missing aircrafts, entire aircrafts, and false alarms, respectively. (a) SAR images of aircraft. (b) Optical images corresponding to SAR targets. (c) Problem of the missing aircrafts and discreteness. (d) Problem of false alarms.

used in SAR image object detection to improve efficiency. Toward the target structure, He *et al.* [10] divided the entire aircraft target into two components constructed by two parallel networks and, finally, located the whole object. However, this algorithm needs to match the corresponding components and is limited by complicated labeling. Zhao *et al.* [25] combined a dilated block with a convolution block attention module to highlight the significant information in fine-grained features. Different from the above anchor-based detectors, the anchor-free methods eliminate the requirement of anchor boxes and gradually become the mainstream development trend of target detection. Cui *et al.* [26] introduced a spatial attention module based on an anchor-free method to enhance the semantic features. In order to further improve the accuracy of small-scale objects, Fu *et al.* [27] put forward an attention pyramid to integrate the information and balance the multiple features from different layers.

Although the above detectors based on deep learning have a certain sound effect in terms of the detection accuracy in the SAR images, there are still some problems. First, due to the imaging mechanism of SAR images and the structural characteristics of aircrafts, a single target is presented as a collection of discrete or isolated strong backscattering points, as shown in Fig. 1(a). The discontinuity of scattering points is easy to cause multiple individual detection results for one target, as indicated in the orange circle. It brings a great challenge to the integrity of aircraft detection. Second, as the imaging condition changes, such as the incident angle and azimuth angle, the same target will result in different imaging results, which increases the complexity of aircraft detection. Especially under certain conditions, the texture and structure of the targets are not prominent, as shown by the yellow circles in Fig. 1(c). Coupled with the interference of background clutter, this will probably lead to the missing detection. Third, due to the limitation of the receptive field, the local information of objects in SAR images is available, but the global structure information is neglected. Similar scatterings caused by the terminal and other buildings at the airport are detected as the aircrafts, which leads to many false alarms in the aircraft detection, as shown in Fig. 1(d). Besides, the strong scattering points of the buildings distributed around the targets are easily confused with aircraft components, inducing the inaccurate position. In addition, the anchor-based detectors inevitably face the optimization of hyperparameters, which is very time-consuming. Hence, these factors mentioned above prove that aircraft detection in the SAR images is a challenging task.

To address these problems, an anchor-free detector based on the scattering feature relation network (SFR-Net) is proposed in this article for aircraft detection in complex SAR images. Different from the methods of directly applying the deep-network model to the SAR images, SFR-Net strengthens the structural relationship among the scattering points and suppresses the clutter through the scattering characteristics. First, SFR-Net takes advantage of the SAR image scattering representations to extract geometric features. For aircraft targets, the strong scattering points are often distributed on several general components, such as the nose, the fuselage, the tail, the wings, and the engines. Our network initially predicts the position of important scattering points in each target. Then, it constructs the structural and spatial relationships among the prime key points and integrates the geometric position feature into the original feature map. Second, the proposed method refines the features between different layers in the pyramid to eliminate the clutter interference, which tackles the unobvious visual scattering in some aircrafts. The merged feature can absorb the beneficial information representations to supplement the original feature. Third, equipped with the ability to improve the relevance of surrounding pixels, our method can obtain the integrated features with the large receptive field, which can be utilized for further identification and regression. Furthermore, unlike most anchor-based approaches, SFR-Net adopts the anchor-free frame and significantly reduces the hyperparameters.

The main contributions of our work can be summarized as follows.

1) The proposed SFR-Net combines the SAR imaging mechanism to comprehensively analyze the distribution of scattering points in the target and the surrounding information in the background. This anchor-free method achieves state-of-the-art performance on the SAR aircraft detection dataset (AIRD).
2) For the individual SAR aircraft, an innovative scattering point relation module (SPRM) is adopted to extract the characteristics and develop the connections among scattering points. This design solves the discreteness problem of aircrafts.
3) In order to strengthen the saliency of the targets under different imaging conditions and reduce the missing aircrafts, a salient fusion module (SFM) is designed to combine the high-resolution detail information with the rich semantic feature, which can enhance the distinguishable features in complex scenes.

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4) Considering the influence of surrounding background interference, we propose a contextual feature attention (CFA) module to capture significant global information from the whole SAR image adaptively and reduce the false alarm in an effective way.

The satisfactory results of ablation experiments demonstrate the effectiveness of the proposed method. The remainder of this article is organized as follows. Section II represents a review of the related work. Section III provides a detailed description of the proposed method for aircraft detection. In Section IV, we evaluate our model on AIRD and analyze the performance of our model. Finally, the conclusions are drawn in Section V.

II. Related Work

This section introduces some object detection methods based on deep learning in the natural scene, the developments of feature pyramids, and some works of object detection in SAR images, respectively.

A. Deep-Learning-Based Object Detectors

Deep learning technology has achieved remarkable success in object detection due to its powerful feature learning ability. The current object detection methods based on deep CNNs include anchor-based and anchor-free detectors.

The anchor-based detectors consist of two- and one-stage methods. Inspired by the idea of region proposals, the Faster R-CNN [28] puts two-stage anchor-based detectors in a dominant position. After that, Cascade R-CNN [29] is presented to improve detection performance by using a cascade structure with different thresholds. Based on the R-CNN framework, some algorithms [30]–[34] leverage the attention mechanism and contextual refinement to extract more representative features. Considering the computational efficiency, one-stage anchor-based detectors, such as YOLOv3 [35], SSD [36], and RetinaNet [37], have attracted much attention because of the faster calculation speed. These methods depend on different level anchors to locate the object more accurately with feature enrichment. Despite the decent performance, the anchor-based detectors require the artificial setting of a large number of parameters, and some objects fail to match the anchor appropriately due to the discrete anchor scales.

Unlike the anchor-based detectors, the anchor-free detectors [38], [39] completely abandon the anchor concept without region proposals, which can fall into keypoint- and center-based methods. For instance, CornerNet [40] and ExtremeNet [41] based on keypoints can directly locate several key points to generate bounding boxes. Reppoints [42] based on offset learning can extract features adaptively to emphasize significant local areas. In terms of center-based methods, FCOS [43] treats all the locations inside the bounding box as positives in a per-pixel prediction manner and predicts four distances to the boundary of the bounding box. Based on DenseBox [44], FoveaBox [45] focuses on the center of the vision field with the highest visual acuity to improve detection performance. At present, the anchor-free methods can accomplish comparable results in speed and accuracy with anchor-based methods.

B. Feature Pyramid Networks in Object Detectors

The feature pyramid network (FPN) [46] is adopted to solve the problem of multiscale object detection. It utilizes the lateral connections through the top-down pathway to combine the shallow layer with high-resolution information and the deep layer with rich semantic information, which can improve the accuracy of multiscale object detection. Compared with FPN, PANet [47] adds another bottom-up pathway to augment the information. The excellent performance of PANet proves the effectiveness of two-way fusion. In the subsequence, NAS-FPN [48], Auto-FPN [49], and EfficientDet [50] achieve more complex bidirectional fusion. NAS-FPN and Auto-FPN leverage neural architecture search to find the optimal FPN structure. By repeatedly stacking multiple weighted feature fusion blocks, EfficientDet obtains an improvement in terms of accuracy. Aiming at the overall performance of the model, Libra R-CNN [51] reduces the imbalance in the sample level, the feature level, and the objective level. To enhance the feature of objects, F^3Net [52] uses a selective fusion strategy to suppress redundant information. Liu *et al.* [53] proposed two complementary modules based on pooling, which can accurately capture the exact position of the objects while sharpening the details of the targets. Besides, NETNet [54] emphasizes the features of small objects by designing an adjacent erasing module and highlights large objects in deep layers by transferring the erased features. These aforementioned methods have proved that multiscale feature fusion is effective in detecting objects. Inspired by it, our method aggregates the features from different layers with rich semantic information and plentiful details to highlight the significant objects.

C. Deep-Learning-Based Object Detection in SAR Images

Different from the traditional target detection methods in SAR images, deep CNNs can automatically extract features without manual design. Deep learning plays an important role in object detection in SAR images because of its satisfactory performance. Chen *et al.* [55] abandoned fully connected layers and presented a new all-convolutional network, which is built by sparsely connected layers. Combined with the shape of the aircraft targets, Fu *et al.* [56] proposed an aircraft target reconstruction method through accurately extracting contour shape features. Based on transfer learning methods, Li *et al.* [57] and Wang *et al.* [58] applied knowledge learned in other fields or tasks to the existed problems of SAR images. Li *et al.* [57] proposed a feature fusion model based on Faster R-CNN to locate the targets in SAR images and avoid overfitting through hard negative mining. Wang *et al.* [58] transferred the parameters obtained by the pretrained network model to the three-channel subaperture SAR images.

In addition, some works focus on the salient features and the attention mechanism. Zhao *et al.* [9] introduced a frequency domain visual attention model based on the pulse cosine transformation to enhance the spatial information for refining bounding boxes. Aiming at the problem of small objects, Jiao *et al.* [59] put forward a dense connection to fuse the multiscale features of different layers with more

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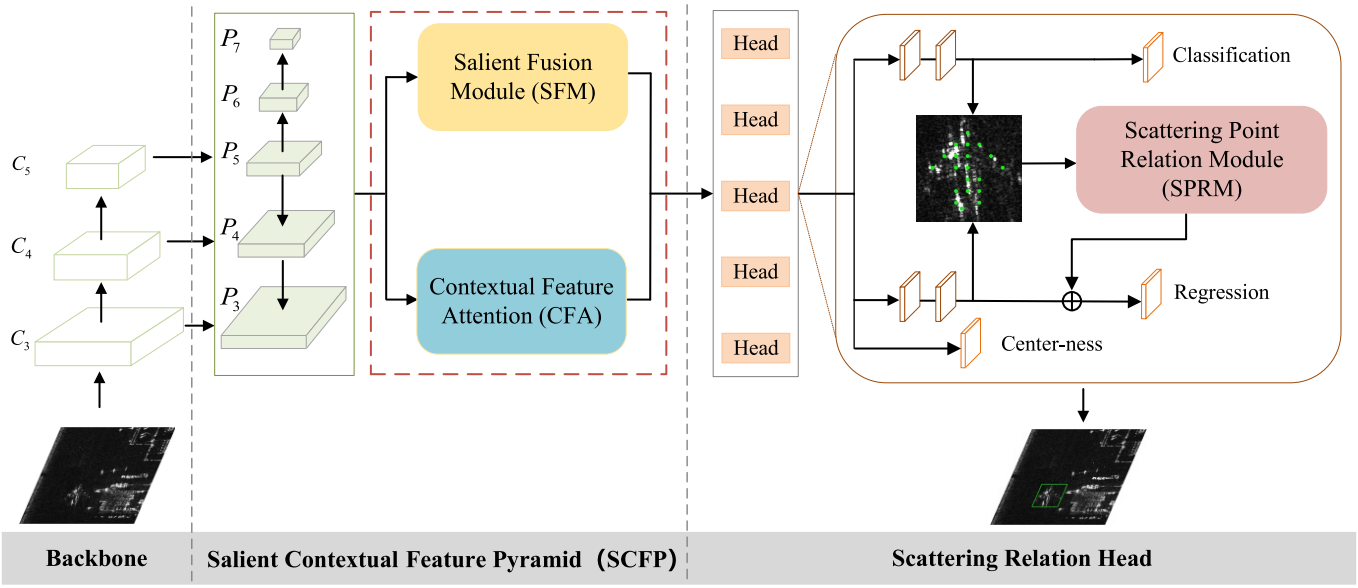


Fig. 2. Overall framework of the proposed SFR-Net. This algorithm includes the backbone, the SCFP, and the scattering relation detection head.

semantic information. Although these methods have achieved good performance, they ignore the scattering characteristics of SAR images. Compared with them, the proposed method consolidates the geometric characteristics of the targets in the SAR images, which can improve the performance of aircraft detection.

III. Methodology

The overall framework of the proposed network SFR-Net is illustrated in Fig. 2, which consists of three parts: the backbone network for feature extraction, the salient contextual feature pyramid (SCFP) for feature fusion, and the scattering relation head for the final prediction. In this section, we first illustrate the common anchor-free baseline. Next, combining the scattering characteristics, the SPRM in the detection head is introduced in detail. Then, we describe the design of two components in the SCFP module. The SFM and CFA complement the valuable information between different layers and enhance the target significance. Finally, the loss function is presented and expressed.

A. Anchor-Free Foundation

At present, most target detectors extract features from the images and then predict the possible location and category of the targets. Compared with the anchor-based methods, the anchor-free detectors reduce the number of hyperparameters, such as artificial anchors and thresholds, which are computationally simpler. FCOS [43] is regarded as an anchor-free method and has many advantages when applied to SAR aircraft detection. First, it leverages each pixel point inside the ground-truth boxes to train the network, which increases the number of positive samples and promotes the balance of positive and negative samples. In addition, aiming at the diversity of aircraft sizes in SAR images, the feature pyramid is constructed to predict objects at different levels. Finally,

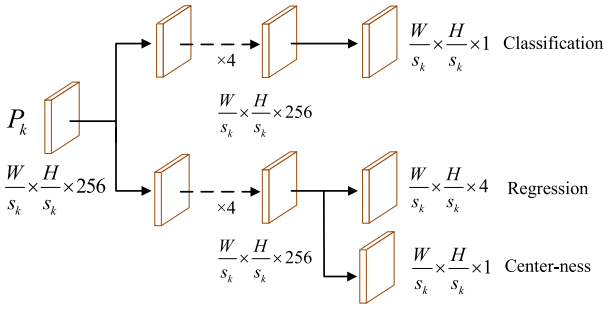


Fig. 3. Structure of the detection head. It contains the classification, regression, and centerness branches.

it presents a novel centerness branch to suppress the inaccurate detected bounding boxes.

In the process of feature extraction, we assume that $P \in \mathbb{R}^{W\times H\times 3}$ is the input image with width W and height H. Through the downsampling ratio $s_k = 2^k$ corresponding to the input image at different layer $k(k = 3, 4, 5, 6, 7)$, the feature maps $P_k \in \mathbb{R}^{(W/s_k)\times(H/s_k)\times C}$ are obtained, where $C = 256$. In order to deal with the objects of different scales, we utilize the features of different layers for detection. Therefore, multiheads are needed to match the characteristics of different layers. The detailed implementation principle of each head is illustrated in Fig. 3. There is a total of three branches arranged with four consecutive 3×3 convolutional layers. In addition to the two branches of binary classification and position regression in the general detection methods, another centerness branch is added. The centerness branch is parallel to the regression branch to calculate the normalized distance from a specific position to the center of the target.

On the feature map P_k in layer k, there exists a pixel point p_{xy}^k at each location (x, y), where $x = 0, 1, \ldots, W/s_k - 1$ and $y = 0, 1, \ldots, H/s_k - 1$. p_{xy}^k corresponds to the position

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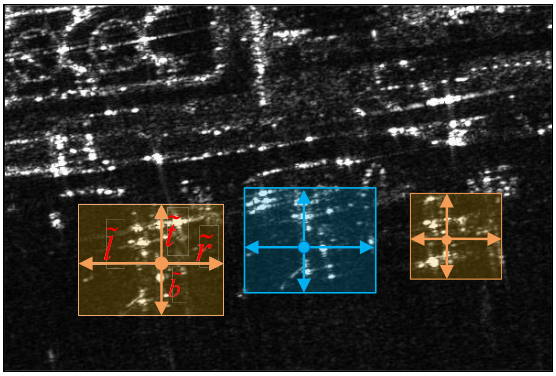


Fig. 4. Encoded location of the bounding box.

$(\dot{x}, \dot{y})$ on the input image, and the mapping relationships are $\dot{x} = xs_k + 0.5s_k$ and $\dot{y} = ys_k + 0.5s_k$. We define the ground-truth bounding box as $B = (x_0, y_0, x_1, y_1, c)$, where (x_0, y_0) and (x_1, y_1) represent the coordinates of the left-top and right-bottom corners, and c is the class of the object, such as aircraft. If the mapping point $(\dot{x}, \dot{y})$ lands in a ground-truth bounding box B, we regard the corresponding pixel point p_{xy}^k as a positive sample and set its class label as the class of the ground-truth box $\tilde{c} = 1$. Otherwise this pixel point is considered as a negative sample and the class label $\tilde{c} = 0$. As for the location regression for the targets, we define the predicted bounding box as a 4-D real vector $\tilde{\boldsymbol{u}} = (\tilde{l}, \tilde{t}, \tilde{r}, \tilde{b})$, where $\tilde{l}, \tilde{t}, \tilde{r}$, and $\tilde{b}$ are denoted as the distances from the location to the four different sides of the bounding box, as illustrated in Fig. 4. For a specific location $(\dot{x}, \dot{y})$ that belongs to bounding box B, its training regression target can be calculated as

$$\begin{aligned} \tilde{l} &= \dot{x} - x_0, \quad \tilde{t} = \dot{y} - y_0 \\ \tilde{r} &= x_1 - \dot{x}, \quad \tilde{b} = y_1 - \dot{y}. \end{aligned} \tag{1}$$

Different scales in the feature pyramid are responsible for detecting the objects with specific scales. We define the maximum distance to regress for layer k as m_k, where m_2, m_3, m_4, m_5, m_6, and m_7 are set as 0, 64, 128, 256, 512, and ∞, respectively. Thence, $[m_{k-1}, m_k]$ is denoted as the scale range to regress a bounding box for layer k. If $\max(\tilde{l}, \tilde{t}, \tilde{r}, \tilde{b})$ for a pixel point is not in the range $[m_{k-1}, m_k]$, this location can be regarded as a negative sample. Considering that a location may be assigned to multiple objects because of the overlapping boxes, we can select a ground-truth box with the minimal area as the corresponding regression target. The detection heads share the parameters to reduce the computational complexity. Moreover, we leverage an exponential function with the trainable parameter to adapt to different layers for better performance.

B. Scattering Point Relation Module

Faced with the incompleteness of the detection targets and inspired by the work [2], [60], we leverage the position information of the strong scattering points to improve the associations between them.

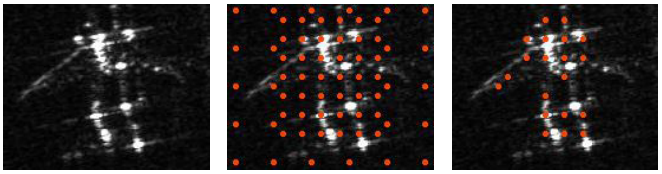


Fig. 5. Visualization of positive sample points. (a) SAR image of an aircraft. (b) All positive sample points. (c) PSPs.

1) Scattering Point Prediction: In the SAR imaging process, the aircrafts present a variety of scattering patterns affected by the differences of the incident angles and azimuth angles. Although aircrafts have different styles and shapes, most of them contain several parts, such as noses, tails, wings, and engines. These components of the aircrafts can easily form a large number of dihedral and trihedral structures, causing strong scattering echoes and relatively large radar cross-sectional (RCS) areas, which reflects the scattering characteristics of ground targets on radar waves. In general, when the scattering is stronger, the RCS is larger, and the area in the image is brighter. Therefore, we can locate the important parts of the aircrafts through the strong scattering points.

We add a novel scattering point prediction branch from the classification branch to predict the confidence of becoming salient points, as shown in Fig. 2. The high confidence in significance can increase the probability of becoming a salient point. We define the top $N(N = 20)$ points with the highest salient confidences among all predicted points of a target as the prime strong scattering points (PSPs) and the scattering label $\tilde{s} = 1$. Meanwhile, the coordinates on the input images of PSPs can also be calculated for the purpose of enhancing the position information in the next step. In order to distinguish the scattering points of different objects, we initially locate the target after the two convolutional layers of the regression branch.

The ground-truth PSPs are denoted as the 20 points with the highest brightness scores among all positive samples of a target. As mentioned in Section III-A, the positive samples contain elements from the different layers of the feature pyramid, as shown in Fig. 5. It turns out that these PSPs can embody the overall outline and geometric features of the target. Considering the influence of speckle noise points, we first carry out 5×5 mean filtering on the original image. Next, the brightness score of a positive sample is calculated in the 7×7 neighborhood in the original image position mapped back from this point. Finally, we sort the brightness scores and select the points corresponding to the top-20 scores as ground-truth PSPs. By embedding the positional relationship among these strong scattering points, we can locate the aircrafts more accurately and avoid the discreteness of the targets.

2) Scattering Point Relation Extraction: Based on the scattering mechanism, the scattering intensity is not obvious and relatively low in some places where there is no dihedral object, such as smooth junction. Therefore, the nose and tail are often separated from the fuselage in SAR images, and the entire aircraft can be easily detected as several individual targets

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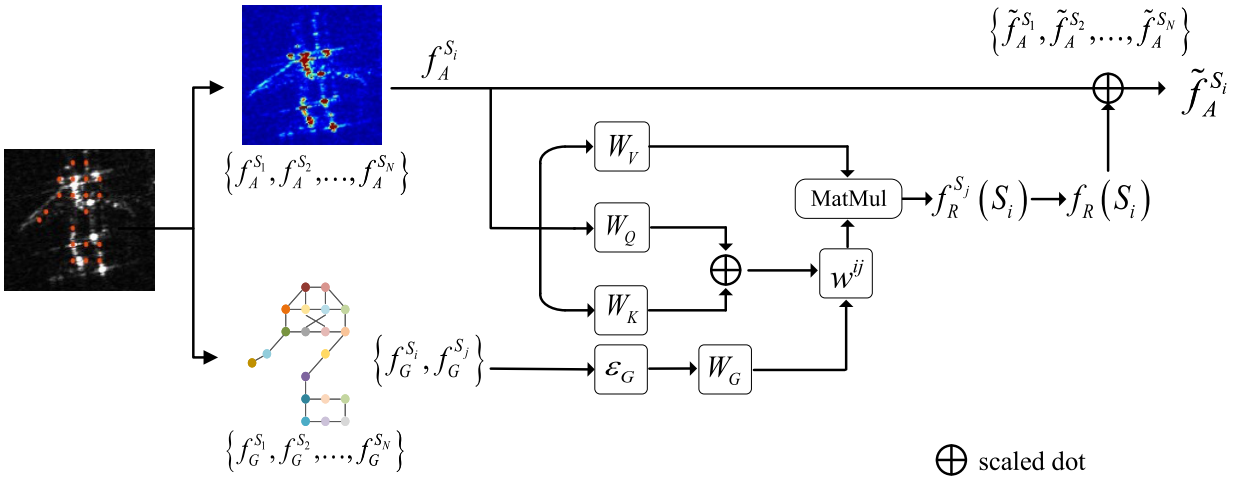


Fig. 6. Structure of the SPRM. f_A represents the appearance feature, and f_G represents the geometric feature.

in the deep learning methods. Hu *et al.* [61] proposed an attention module improved from the self-attention mechanism to model the pairwise relations between objects. However, this module contains RoI pooling, and its inputs are the predicted proposals, which is suitable for two-stage methods with anchors. Taking the different goals and applications into consideration, our basic idea aims to enhance the relationships among points in an anchor-free method instead of proposals.

In order to describe the connections between important PSPs, we propose an SPRM structure, which is shown in Fig. 6. Let $S = \{S_1, S_2, S_3, \ldots, S_N, N = 20\}$ denote a set of PSPs for every object in which each point is represented by 2-D coordinates. Let a PSP S_i consist of the appearance feature $f_A^{S_i}$ and geometric feature $f_G^{S_i}$, where $f_A^{S_i}$ is the feature directly extracting from corresponding coordinate position on feature map P_k and $f_G^{S_i}$ is transformed from a simply 2-D point with its coordinate (x_i, y_i).

We consider the self-attention as the operation that reflects interactions between these points. In the scaled dot-product attention [60], if the input consists of query, key, and value packed with the representations, Q, K, and V, respectively, the output value can be compactly computed using matrix multiplications

$$v^{\text{out}} = \underbrace{\text{softmax}\left(\frac{QK^t}{\sqrt{d_k}}\right)}_{\text{attention weight}} V \tag{2}$$

where Q and K compile with dimension d_k that can be regarded as a scaling factor on the denominator. In our work, the total feature set of N scattering points is denoted as $\{(f_A^{S_i}, f_G^{S_i})\}_{i=1}^N$. We calculate the relation feature between the ith point S_i and various PSPs through the expression as

$$f_R(S_i) = \sum_j w^{ij} \cdot \left(W_V \cdot f_A^{S_j}\right). \tag{3}$$

The output f_R is obtained by the weighted appearance features from other points through linear transformation W_V, and w^{ij} represents the relation weight that reflects the appearance feature and geometric feature of different points. This process is completed by matrix multiplication, which is represented as MatMul in Fig. 6. We define the appearance weight and geometry weight as w_A^{ij}, w_G^{ij}, respectively. The w_A^{ij} is computed as

$$w_A^{ij} = \frac{\text{dot}\left(W_K f_A^{S_i}, W_Q f_A^{S_j}\right)}{\sqrt{d_k}} \tag{4}$$

where W_K and W_Q are the appropriate matrix parameters that can be learned adaptively during the training process, similarly as (2).

Unlike the general purpose of predicting the bounding boxes, we encode geometric features with 2-D coordinates instead of the 4-D information feature. The 2-D relative f_G is computed as $(\log(|x_i - x_j|), \log(|y_i - y_j|))^T$, which can keep scale transformation and translation invariant, and the logarithmic function can avoid the extensive range of numerical changes.

Inspired by the sine and cosine functions of different frequencies in [60], we convert the 2-D features into high-dimensional embedding representations, denoted as ε_G. Then, W_G is added to transform the embedded feature into a scalar. Thus, the geometry weight is computed as

$$w_G^{ij} = \text{ReLU}\left\{W_G \cdot \varepsilon_G\left(f_G^{S_i}, f_G^{S_j}\right)\right\}. \tag{5}$$

The ReLU function is utilized to trim at zero nonlinearity. Considering the different influences of other PSPs on point S_i, we combine appearance and geometry weight to define w^{ij} as

$$w^{ij} = \frac{w_G^{ij} \cdot \exp\left(w_A^{ij}\right)}{\sum_{j=1}^N w_G^{ij} \cdot \exp\left(w_A^{ij}\right)}. \tag{6}$$

When the relation weight w^{ij} is obtained, the relation features of the point set S can be obtained from (3). Then, we enhance the connections between S_i and other PSPs by concatenating the relation features on the basis of the original appearance feature, which is calculated as follows:

$$\tilde{f}_A^{S_i} = f_A^{S_i} + \text{Concat}\left[f_R^{S_1}(S_i), \ldots, f_R^{S_N}(S_i)\right]. \tag{7}$$

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After we get the feature $\tilde{f}_A$ from the SPRM structure, we perform a concatenation operation to aggregate the $\tilde{f}_A$ with the original feature map P_k for further detection. In SPRM, the geometric feature can effectively augment the relationship among scattering points of aircrafts, which can provide holistic information to avoid the discreteness of objects. In addition, this module can obtain output features of the same dimension as the input, which can be widely used as a superimposed block in any network.

C. Salient Contextual Feature Pyramid

Due to the limitation of convolution kernel size, convolution operation focuses on local information and lacks global information of the whole feature map. It causes the surrounding buildings in the complex background to be easily confused with the aircraft targets during detection. Moreover, the interference of the speckle noise in SAR images causes the aircraft targets to be insufficiently prominent, which increases the difficulty of classification and positioning. Inspired by the work in [62], the SCFP is proposed to reduce the false alarm and improve the localization accuracy, which consists of two subnets: SFM and CFA. The SFM aims to adaptively highlight the significant regions of aircraft objects, and the CFA is introduced to enhance global information at different levels.

1) Salient Fusion Module: In this method, the ResNet-50 [63] is utilized as the backbone to acquire image representative features in different levels. The shallow convolutional layer in the backbone is prone to extract low-level features, such as edges, textures, and shapes. When the levels continue to deepen, more abstract and distinguishable features can be obtained. As shown in Fig. 2, the last three layers of the backbone in the bottom-up path are denoted by $\{C_3, C_4, C_5\}$. Then, the high-level features are upsampled through nearest-neighbor upsampling and integrated by the previous layer through the lateral connection to generate $\{P_3, P_4, P_5\}$ in the top-down direction. This process is introduced as FPN [46] and is calculated as

$$I_k = \begin{cases} \text{Conv}_{1\times1}(C_k), & k = 5 \\ \text{Upsample}(I_{k+1}) + \text{Conv}_{1\times1}(C_k), & k = 3, 4 \end{cases} \tag{8}$$

$$P_k = \begin{cases} \text{Conv}_{3\times3}(I_k), & k = 3, 4, 5 \\ \text{Conv}_{3\times3}(P_{k-1}), & k = 6, 7 \end{cases} \tag{9}$$

where $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ represent the 1×1 and 3×3 convolution, and $I_k \in \mathbb{R}^{(W/s_k)\times(H/s_k)\times C}$ refers to the converged features. Furthermore, P_6 and P_7 are produced by the P_5 and P_6 undergoing with a 3×3 convolution with a stride of 2, respectively. The high-level features through many convolutional layers contain more semantic information with the deficiency of accurate location information, while the low-level features involve plentiful details and background noise. Thus, the SFM is presented to refine the different level features to eliminate the intervention of clutter. Considering the distribution layer of the positive samples, there are few samples in P_6 and P_7 due to the small size of these two feature maps. As for the feature map P_3, many positive samples in this layer are located at the edge of the targets with relatively low centerness scores. Thereby, we concentrate on feature maps P_4 and P_5 to emphasize the salient objects, which contains most of the positive samples with good quality.

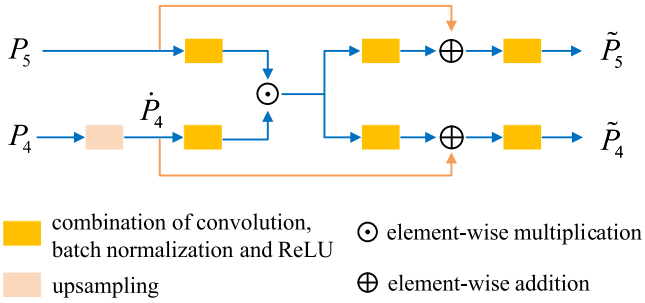


Fig. 7. Structure of the SFM.

Specifically, the SFM is divided into two branches: one is P_4 that is regarded as a low-level feature map, and the other is P_5 for the high level, as shown in Fig. 7. First, P_4 is executed as $\dot{P}_4$ by the double upsampling to maintain the same size as P_5. Then, the 3×3 convolutional layer is carried on $\dot{P}_4$ and P_5, respectively, and their features are merged by elementwise multiplication. The aggregated features combine the advantages of excellent semantics and precise details in two different levels. Finally, the fused features are constructed with the original features to retain more salient information, which can be generated as

$$\dot{P}_4 = \text{Upsample}(P_4) \tag{10}$$

$$\begin{cases} \tilde{P}_5 = U_h, & \left(P_5 + M_h\left(G_l\left(\dot{P}_4\right) \odot \left(G_h(P_5)\right)\right)\right) \\ \tilde{P}_4 = U_l, & \left(\dot{P}_4 + M_l\left(G_l\left(\dot{P}_4\right) \odot \left(G_h(P_5)\right)\right)\right) \end{cases} \tag{11}$$

where $\odot$ is defined as the elementwise multiplication, and each of U_h, U_l, M_h, M_l, G_h, and G_l is composed of the 3×3 convolution, batchnorm, and relu. The output features in SFM are $\tilde{P}_4 \in \mathbb{R}^{(W/2^4)\times(H/2^4)\times C}$ and $\tilde{P}_5 \in \mathbb{R}^{(W/2^5)\times(H/2^5)\times C}$. After obtaining the fused features, we perform the 3×3 convolution on $\tilde{P}_4$ to preserve the same dimensions with P_4. The SFM can absorb the beneficial information representations of P_4 and P_5 to supplement the original feature. The common parts between P_4 and P_5 are highlighted to generate the salient regions by elementwise multiplication, and the speckle noise can be suppressed due to its random distribution.

2) Contextual Feature Attention: To augment the contextual information and reduce inaccurate localization, the CFA is proposed to increase the global characteristics obtained by the pyramid. This process is illustrated in Fig. 8. Given a local feature map $P_k \in \mathbb{R}^{W_k\times H_k\times C}$ $(W_k = W/2^k, H_k = H/2^k)$ at different layers $k(k = 3, 4, 5)$, we first perform three 1×1 convolution on P_k to generate feature maps $Q, K, V \in \mathbb{R}^{W_k\times H_k\times C^Q/C^K/C^V}$, respectively, where C^Q, C^K, and C^V are the different numbers of channels. We define the vector for a special position u in the spatial dimension of Q as $Q_u \in \mathbb{R}^{C^Q}$. To aggregate the feature in the same row and column as position u, we utilize the affinity operation [62] on Q_u and

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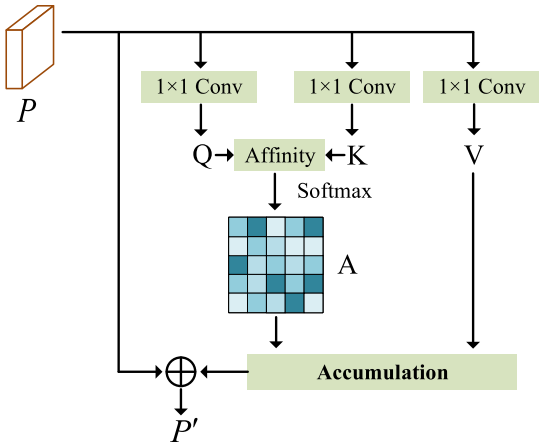


Fig. 8. Structure of the CFA.

$K_u \in \mathbb{R}^{(W_k+H_k-1)\times C^K}$ to calculate their relevance

$$d_{m,u} = Q_u K_{m,u}^T \tag{12}$$

where m represents one of the $W_k + H_k - 1$ members in K_u and $K_{m,u} \in \mathbb{R}^{C^K}$. The collection of $d_{m,u}$ is denoted as $D \in \mathbb{R}^{(W_k+H_k-1)\times W_k \times H_k}$. Afterward, the softmax function is operated on D over the spatial dimension to generate normalized feature map A.

Considering $V \in \mathbb{R}^{W_k \times H_k \times C^V}$ ($C^V = C$), we define the vector in the same row or column with u as $V_u \in \mathbb{R}^{(W_k+H_k-1)\times C^V}$, and $V_{m,u} \in \mathbb{R}^{C^V}$ is the mth elements among V_u. The new feature map P'_u at position u with global information is computed by the accumulation operation

$$P'_u = \sum_{m\in|V_u|} A_{m,u} V_{m,u} + P_u \tag{13}$$

where m is the channel of position u, and $A_{m,u}$ is the corresponding scalar value in A. The global features of the surrounding pixels around position u are aggregated by the mentioned process CFA. The output representation P' equipped with the CFA has the same dimension as the input feature map P.

In terms of aircraft detection, the CFA module can avoid ignoring the background information around the aircrafts that are often parked on the tarmac or appear near the terminal. Combined with the global characteristics, the feature map for classification can increase the distinction from the interference of similar surrounding buildings, which is beneficial to decrease the misclassification of brighter buildings as aircraft targets. In addition to reducing false alarms, this module utilizes more useful contextual information so that it can improve the accuracy of aircraft positioning.

In order to suppress the interference of background while making the targets prominent, we combine SFM and CFA in the feature pyramid, which is shown in Fig. 9. Then, we apply the 3×3 convolution to obtain the final fused features. The aggregated feature can adaptively highlight the salient objects and improve the performance of our network.

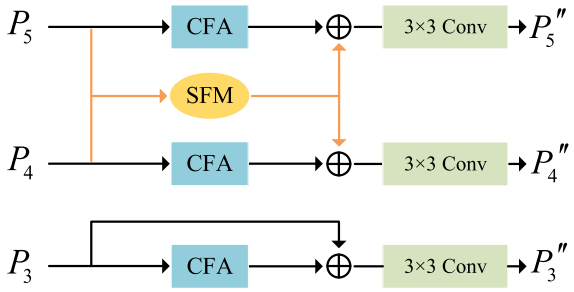


Fig. 9. Overall structure of the SCFP. SFM and CFA are the subnets.

D. Loss Function

The total training loss function can be divided into four parts

$$L = \frac{1}{N_{\text{pos}}} \sum_k \sum_{x,y} \Big\{ L_{\text{cls}}\big(c^k_{xy}, \tilde{c}^k_{xy}\big) + \big[\tilde{c}^k_{xy} = 1\big] L_{\text{loc}}\big(\boldsymbol{u}^k_{xy}, \tilde{\boldsymbol{u}}^k_{xy}\big) + L_{\text{point}}\big(s^k_{xy}, \tilde{s}^k_{xy}\big) + L_{\text{cen}} \Big\} \tag{14}$$

where L_{cls}, L_{loc}, L_{point}, and L_{cen} refer to the classification loss, localization loss, scattering point loss, and centerness loss, respectively. $[\tilde{c}^k_{xy} = 1]$ is the indicator function, and its value is 1 if $\tilde{c}^k_{xy} > 1$; otherwise, it is 0. N_{pos} denotes the number of positive samples.

L_{cls} and L_{point} are derived from focal loss [37], which can adjust the weight of positive and negative samples to ease their imbalance. It can be formulated as

$$\text{FocalLoss}(c_t) = -\mu_t (1 - c_t)^\gamma \log(c_t) \tag{15}$$

where $\mu_t = 0.25$ and $\gamma = 2$ are the focusing parameters. The probability c_t is calculated by

$$c_t = \begin{cases} c^k_{xy}, & \text{if } \tilde{c}^k_{xy} = 1 \\ 1 - c^k_{xy}, & \text{otherwise.} \end{cases} \tag{16}$$

L_{loc} is the GIoU loss [64] to measure the intersection between the predicted bounding boxes and the ground-truth boxes. L_{cen} is the binary cross entropy loss, which can evaluate the degree of difference between the predicted and ground-truth centerness scores. For a position (x, y), the score can be formulated as

$$\text{centerness}^* = \sqrt{\frac{\min(\tilde{l}, \tilde{r})}{\max(\tilde{l}, \tilde{r})} \times \frac{\min(\tilde{t}, \tilde{b})}{\max(\tilde{t}, \tilde{b})}} \tag{17}$$

where $\tilde{l}, \tilde{t}, \tilde{r}$, and $\tilde{b}$ donate the regression targets. At inference phase, the predicted centerness scores are multiplied by the regression scores of the corresponding pixel point to generate the final scores, which can suppress the low-quality bounding boxes away from the center of an object.

IV. Experiments

In this section, we conduct extensive experiments on AIRD from the Gaofen-3 satellite and analyze the effectiveness of the proposed method. First, we introduce the dataset and experimental details. Second, the evaluation metrics are described.

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Then, the ablation experiments are implemented, and the performances of different compositions are analyzed in detail. Finally, we establish the comparison with the CNN-based methods and illustrate the performance in different complex scenes.

A. Dataset and Settings

The AIRD is a large-scale aircraft dataset containing 109 single-polarization SAR images from the Gaofen-3, which is a C-band satellite. These magnitude images with expert annotation cover 17 different airports or bases in multiple complex scenes, which increases the difficulty of detection due to the similar scattering phenomenon of nearby buildings and clutter interference. The large-scale images are about 30 000 × 25 000 pixels with a 1-m resolution, which provides 5357 SAR aircrafts and consists of two different polarization methods (HH and VV). We randomly choose 78 images as the training and validation set, and the remaining 31 images are regarded as the test set. Due to the restriction of GPU memory and processing speed, we avoid the way of training the network on the entire large image. We crop each large SAR image into 600 × 600 pixels with the overlap of 200 pixels in a sliding window mode. A total of 4429 aircraft slices are obtained, and the criteria are calculated on slices in the inference phase. The AIRD is established to conduct the experiments to demonstrate the effectiveness of the method.

The backbone of the proposed method is the ResNet-50 [63], which is pretrained on the ImageNet. The convolutional layers in our method are initialized in the same way as [37]. We utilize the stochastic gradient descent (SGD) [48] with a batchsize of four images to train this model. The momentum and weight decay of the optimizer are set to 0.9 and 0.0001, respectively. We set the initial learning rate to 0.001 and decay it to 0.0001 after 40 epochs for the convergence of loss. The proposed method is implemented on the PyTorch 1.2 framework, CUDA 10.0, and performed on NVIDIA RTX 2080 Ti GPU.

B. Evaluation Metric

In order to evaluate the performance of the aircraft detection task, we adopt some widely used metrics: precision, recall, F1-score, and average precision (AP). Precision and recall represent the detection accuracy and completion rate of the detector, respectively, which are computed as follows:

$$\text{Precision} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{18}$$

$$\text{Recall} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{19}$$

where N_{TP} is the number of true positives (TPs) that refer to the correctly detected objects. N_{FP} denotes the number of false positives (FPs) that represent false alarms. N_{FN} is the number of false negatives (FNs) that indicate missing aircrafts.

The intersection over union (IoU) denotes the degree of overlap between the predicted bounding box and the ground-truth bounding box in the images. The value of IoU ranges from 0 to 1, and its value is 1 when the two bounding boxes completely overlap. Generally, if the IoU of a detected bounding box with its corresponding ground truth is greater than 0.5, this box can be regarded as a TP. Besides, we leverage F1 to reflect the comprehensive performance of the models, which is defined as

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{20}$$

There are different precision and recall under different confidence thresholds, and we can draw a 2-D curve with the recall on the abscissa and the precision on the ordinate, which is called the PR curve. The AP metric is defined as the area under the precision–recall curve, and it is used to assess the overall quality of a classifier, which is computed as

$$\text{AP} = \int_0^1 P(R) dR. \tag{21}$$

In this article, $\text{AP}_{0.5}$ is an important metric that is calculated at IoU = 0.5. In general, the AP value and detector performance are positively correlated. In order to more strictly reflect the accuracy of positioning, we add two other indicators $\text{AP}_{0.6}$ and $\text{AP}_{0.7}$, which are obtained at IoU = 0.6 and IoU = 0.7, respectively.

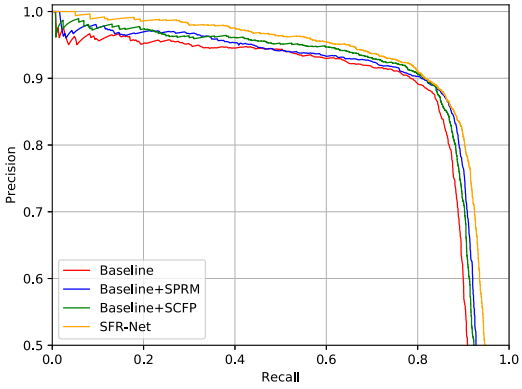


Fig. 10. PR curves of different improvements in SFR-Net.

C. Ablation Studies

In this section, we conduct a series of ablation experiments on AIRD to analyze the effectiveness of each proposed component in SFR-Net. All subsequent experiments maintain the same settings to make a fair comparison. The specific comparisons of the experimental results are displayed in Table I. The precision and recall at different thresholds in the PR curve are presented in Fig. 10. It can be seen that different designed components have improved the results to a certain extent. Since the fused features in SFR-Net contain more semantic and location structure information, the aircraft targets can be positioned more accurately. As shown in Table I, compared with the baseline (FCOS), $\text{AP}_{0.5}$ and $\text{AP}_{0.7}$ of the proposed method can achieve 3.5% and 3.1% improvements when the IoU is set as 0.5 and 0.7, respectively. In addition, the two components have achieved some degrees of improvement in detection accuracy, while the model parameters (Params) and floating-point operations (FLOPs) are only slightly increased.

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TABLE I
PERFORMANCE OF EACH COMPONENT IN THE PROPOSED SFR-NET ON AIRD

SPRM	SCFP	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.6}$	$AP_{0.7}$	Params(M)	FLOPs(G)
×	×	0.8749	0.8294	0.8515	0.8570	0.7426	0.4829	**30.84**	**71.51**
✓	×	0.8816	0.8498	0.8654	0.8814	0.7683	0.5084	33.02	71.53
×	✓	0.8858	0.8438	0.8643	0.8813	0.7671	0.5061	37.91	87.07
✓	✓	**0.8905**	**0.8567**	**0.8733**	**0.8923**	**0.7768**	**0.5143**	39.09	87.09

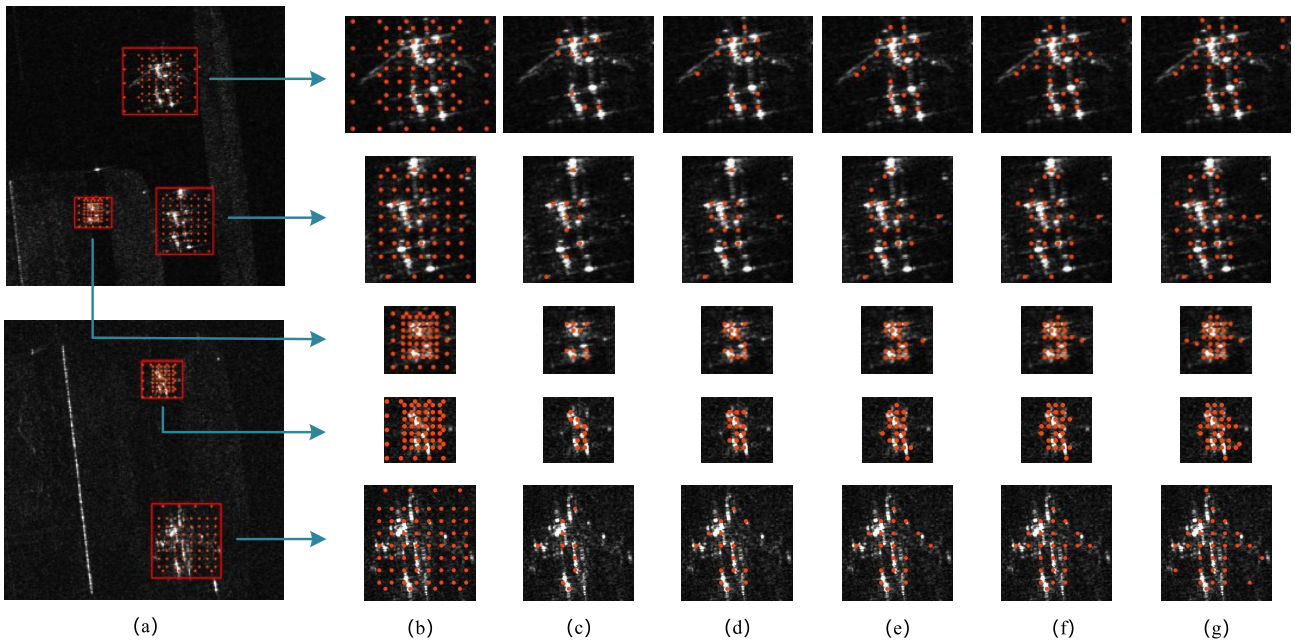


Fig. 11. Results of different numbers of PSPs. (a) Ground truths and positive samples. (b) All positive samples for each aircraft. (c)–(g) Results of 10, 15, 20, 25, and 30 scattering points, respectively.

TABLE II
EFFECT OF DIFFERENT NUMBERS OF PSPS

the number of PSPs	Precision	Recall	F1	$AP_{0.5}$
10	0.8476	0.8235	0.8354	0.8661
15	0.8577	0.8357	0.8466	0.8697
20	**0.8816**	**0.8498**	**0.8654**	**0.8814**
25	0.8792	0.8457	0.8621	0.8753
30	0.8720	0.8442	0.8579	0.8712

Next, we will make a detailed experimental analysis of the SPRM, including SAR scattering characteristics. Furthermore, the SFM and CFA in the SCFP are evaluated separately for their performance.

1) Effect of SPRM: We first analyze the number of PSPs, where the geometric characteristics of the relationship among PSPs are integrated behind the two convolutional layers in the regression branch. We select different numbers of PSPs (10, 15, 20, 25, and 30) for each aircraft target with multiple scales for comparative experiments, and the results are shown in Table II and Fig. 11. It can be seen that $AP_{0.5}$ and other indicators are optimal when the selected number is designated as 20.

Through the statistical analysis of aircraft targets, the distributions of diagonal length and target size are shown in Fig. 12. It is found that the aircrafts exhibit a multiscale distribution, and the diagonals of most aircrafts are mainly concentrated in the range of 40–80 m. When fewer PSPs are selected, the position features between scattering points are difficult to be comprehensively extracted, and the structural contour features of the aircrafts are not clear enough. However, a large number of PSPs in the small aircrafts tend to be excessively clustered together, which leads to ignorance of the positional differences between the scattering points. Taking the various sizes of the aircrafts into account, we extract 20 PSPs from each object to explore the connection between them, which achieves a good balance between the ability to reflect geometric characteristics and the computational complexity.

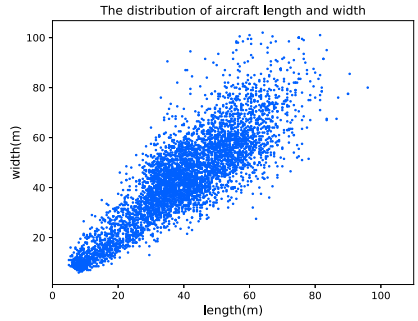
We analyze the performance when SPRM is added to the different convolutional layers of the regression branch, and the experimental results are listed in Table III. It can be seen that the position of SPRM between the convolutional layers makes little difference, so we choose to arrange the SPRM after two convolutional layers, which is the optimal choice with better performance.

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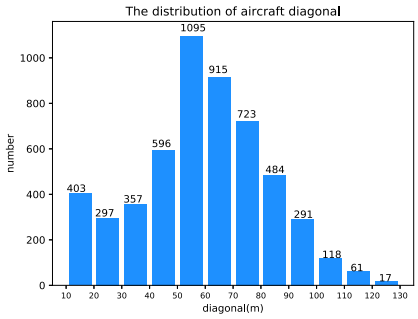
TABLE III

INFLUENCE OF CHANGING THE NUMBER OF CONVOLUTIONAL LAYERS BEFORE SPRM

#conv	Precision	Recall	F1	$AP_{0.5}$
1	0.8749	0.8257	0.8496	0.8731
2	**0.8816**	**0.8498**	**0.8654**	**0.8814**
3	0.8612	0.8209	0.8406	0.8786



(a)



(b)

Fig. 12. (a) Distribution of aircraft length and width. (b) Distribution of aircraft diagonal.

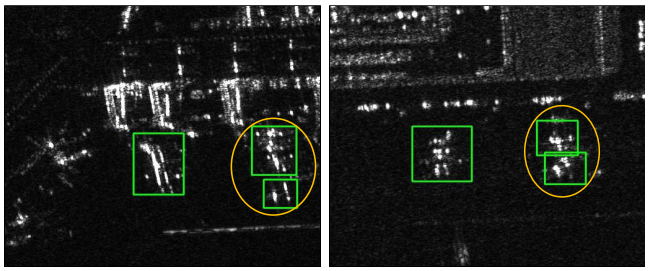
TABLE IV

ABLATION STUDIES OF SFM AND CFA

SFM	CFA	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.6}$	$AP_{0.7}$
×	×	0.8749	0.8294	0.8515	0.8570	0.7426	0.4829
✓	×	0.8807	0.8404	0.8601	0.8709	0.7578	0.4957
×	✓	0.8813	0.8397	0.8600	0.8762	0.7594	0.4986
✓	✓	**0.8858**	**0.8438**	**0.8643**	**0.8813**	**0.7671**	**0.5061**

As mentioned in Section III-B, we add the geometric features of the positions between the scattering points to detect aircraft targets. The second and third rows in Table I show that the network with SPRM increases the $AP_{0.5}$ and $AP_{0.7}$ by 2.4% and 2.5% over baseline, respectively. In addition, the detection results in Fig. 13 illustrate its effective role in resisting the discreteness of targets. The baseline method in Fig. 13(a) detects the aircraft nose and fuselage as two separated objects, while the proposed SPRM method in Fig. 13(b) can maintain the complete contour structure of the aircrafts.

2) Effect of SFM: Table IV demonstrates the effects of two subnets in the SCFP by gradually adding them to the



(a)



(b)

Fig. 13. Detection results of the baseline without and with SPRM. The green boxes and orange circles represent detection results and the entire aircrafts, respectively. (a) Aircraft detection results of the baseline without SPRM. (b) Aircraft detection results of the baseline with SPRM.

baseline. We first analyze the impact of SFM from the first three rows of Table IV. It can be seen that both $AP_{0.5}$ and recall gain the improvements. The proposed SFM complements the important features of different layers by integrating high-level semantic features and low-level positioning information, so as to highlight the features of important locations and reduce the missing objects. The comparison of the detection results is shown in Fig. 14. For instance, we use the confidence scores of the last layer in the classification branch to draw the visualization of feature maps. We can observe that the baseline may ignore some unobvious targets in Fig. 14(a) and (c), whose corresponding visualization results are demonstrated in Fig. 14(b) and (d). In contrast, the proposed SFM can make the objects prominent and enhance the saliency of the objects, as shown in Fig. 14(e) and (g), which improves the ability of the network to suppress the speckle noise in SAR images.

3) Effect of CFA: The results in Table IV show that the CFA module increases $AP_{0.5}$ by 1.9% compared with the baseline. The CFA combines the contextual information from the entire image with global features to more accurately detect the aircrafts. The detection results in Fig. 15 also demonstrate the superiority of the CFA structure. We can observe that some parts of the boarding bridges are wrongly detected as aircrafts due to the similar scattering in Fig. 15(a). In comparison, the CFA can reduce false alarms in the background by associating pixels on a global scale in Fig. 15(b), which brings an improvement in effect. Furthermore, the calculated IoU between the predicted target and the corresponding ground truth has been improved to a certain extent, and the higher $AP_{0.6}$ and $AP_{0.7}$ values also indicate that the localization is more accurate than the baseline.

Up until now, we have separately demonstrated the validity of the two submodules in the designed pyramid. The overall

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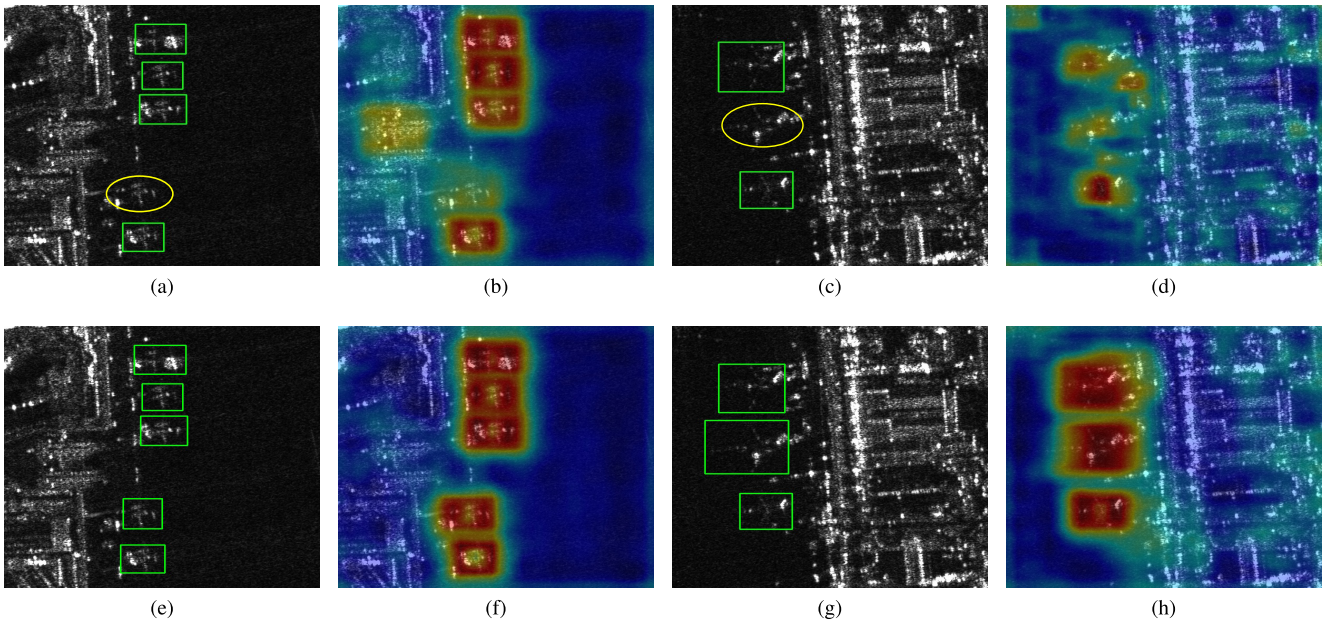


Fig. 14. Detection results of the baseline without and with SFM. The green boxes and yellow circles are the detection objects and the missing aircrafts, respectively. (a) and (c) Detection results of the baseline. (b) and (d) Visualization of corresponding feature maps. (e) and (g) Detection results of the baseline with SFM. (f) and (h) Visualization of corresponding feature maps.

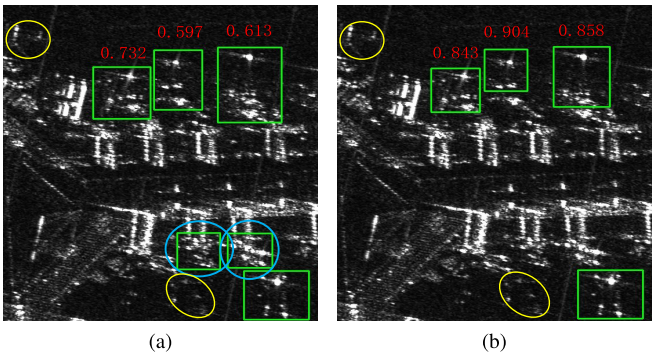


Fig. 15. Detection results of the algorithms without and with CFA. The green boxes, blue circles, and yellow circles represent the detection results, the false alarms, and missing aircrafts, respectively. The red values reflect the IoU between the predicted objects and the corresponding ground truth. (a) Aircraft detection results of the baseline. (b) Aircraft detection results of the baseline with CFA.

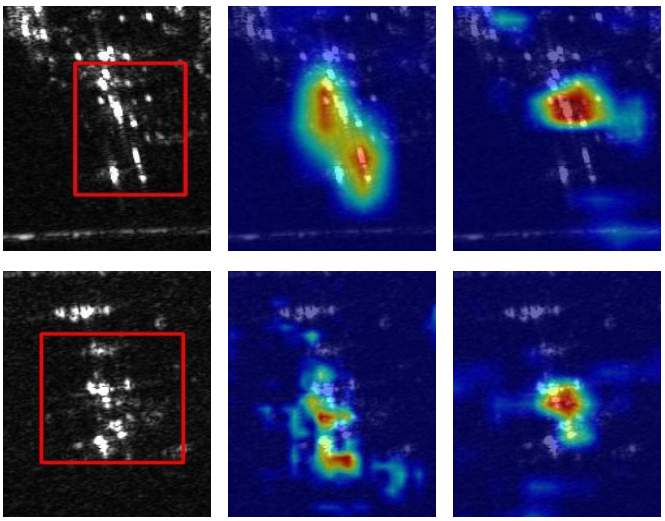


Fig. 16. Comparison visualization of the confidence maps. (a) Ground truth in SAR images. (b) Visualization of the baseline. (c) Visualization of the baseline with SPRM.

results are presented in Table IV. It turns out that the SFM and CFA can not only reduce missing objects and false alarms in complex scenes but also provide precise localization and achieve a better detection performance.

D. Discussion

In this section, we will further discuss and explain the various components and overall effects of SFR-Net. We qualitatively analyze the proposed components to evaluate their effectiveness.

1) SPRM: Considering the scattering characteristics of SAR images, SPRM is utilized to enhance the relationship among prime scattering points, which can deal with the problem of discreteness. The different geometric characteristics in the range and the azimuth directions make the object visually present a state of structural deviation. At some incident angles, the smooth and streamlined parts connected between the metal components reflect the electromagnetic waves weakly and blend with the background. Different echoes and imaging radiation characteristics of various parts lead to the failure of the aircraft to be integrated as a whole in the SAR image, as shown in Fig. 1(a). In addition, due to phenomena such as foreshortening, layover, and shadows, the targets are usually distorted to some extent.

The proposed method compensates for the separation of different components by linking the position characteristics between the scattering points, as indicated in Fig. 16. It turns

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TABLE V
COMPARISON OF DIFFERENT CNN-BASED METHODS ON AIRD

Method	Anchor-free	Precision	Recall	F1	$AP_{0.5}$	$AP_{0.6}$	$AP_{0.7}$	Params(M)	FLOPs(G)
SSD	×	0.8696	0.7924	0.8292	0.8205	0.6986	0.4421	36.10	72.62
Faster R-CNN	×	0.8018	0.8196	0.8106	0.8460	0.7154	0.4586	41.12	82.16
Reppoints	✓	0.8643	0.8225	0.8429	0.8476	0.7294	0.4643	36.60	**67.45**
FoveaBox	✓	0.8561	0.8274	0.8415	0.8489	0.7408	0.4825	**36.01**	71.92
Casacde R-CNN	×	0.8394	0.8291	0.8342	0.8505	0.7452	0.4879	68.93	109.97
PADN	×	0.8672	0.8283	0.8473	0.8604	0.7465	0.4851	45.98	93.97
SFR-Net	✓	**0.8905**	**0.8567**	**0.8733**	**0.8923**	**0.7768**	**0.5143**	39.09	87.09

out that, in the inference process of the baseline, the discrete nose and fuselage are regarded as two separate small aircraft targets. Compared with it, the visualization of the feature maps in Fig. 16(c) reflects the entirety between different components of the targets. As shown in the detection results of Fig. 13 and the visualization of Fig. 16, the designed SPRM integrates the appearance features and geometric features for detection, so as to improve the integrity among aircraft parts, which is conducive to the better positioning of the aircraft.

2) SCFP: The proposed SCFP aims to highlight significant features and capture the contextual information around the target in the complex background. Table IV illustrates the improvement of each component on various indicators. The heatmaps in Fig. 14 intuitively show the visualization results of SFM. We can observe that the yellow circled areas on the matching heatmaps are not obvious, and the targets inside these areas are ignored. In contrast, the corresponding positions of the missing aircrafts on the improved heatmaps in Fig. 14(f) and (h) become more prominent and brighter, which can obtain more attention to detect.

Combined with the global information brought by the CFA module, the false alarm under background interference is reduced, and the positioning is more accurate, which is shown in Fig. 15. However, there are still some cases of missing aircrafts near the terminal, as illustrated by the yellow circles in Fig. 15. This problem is mainly caused by two reasons. First, due to the change in the scattering conditions of the SAR images, such as the incident angle, the imaging of the target is not stable. In some cases, the scattering of the metal components in the aircraft is not obvious enough. Compared with background interference with strong scattering, the semantic information of aircraft is inadequate. The other reason is that the nose part of the aircraft is often integrated into the terminal building, which results in the lack of a complete object character. Therefore, some targets are still arduous to detect after aggregating contextual information and enhancing salience. Aiming at the interference of buildings, such as terminal buildings, fusing imaging mechanisms to maintain accurate detection between close aircraft targets is the focus of our future research direction.

E. Comparison With CNN-Based Methods

To validate the effectiveness of SFR-Net, we compare it with other methods and conduct a series of experiments on AIRD. Considering the comprehensiveness of the categories in CNN methods, we choose six different methods—SSD, Faster R-CNN, Reppoints, FoveaBox, Cascade R-CNN, and pyramid attention dilated network (PADN) [25]—for comparative experiments, which cover the scope of anchor-based and anchor-free methods. The PADN introduces dilated attention block to refine significant information and obtain fine-grained feature maps for aircraft detection. The comparison results with other CNN methods are shown in Table V, and the detailed PR curves are illustrated in Fig. 17. We can observe that SFR-Net has a significant improvement of more than 3.2% and 2.6% on the AP and F1 index, respectively, which also achieves the best performance in terms of precision and recall. In addition, the increased $AP_{0.7}$ verifies that the positioning of bounding boxes is more accurate, which is conducive that the bounding boxes can contain less background and facilitate the identification of the targets in the detection process. Compared with some lightweight detectors (such as SSD, Reppoints, and FoveaBox), although the parameters of our method have increased, the performance has been greatly improved. Thus, our proposed method maintains a certain balance between the computational complexities with the detection accuracy. Furthermore, the PR curves also confirm the appreciable improvement of the proposed method.

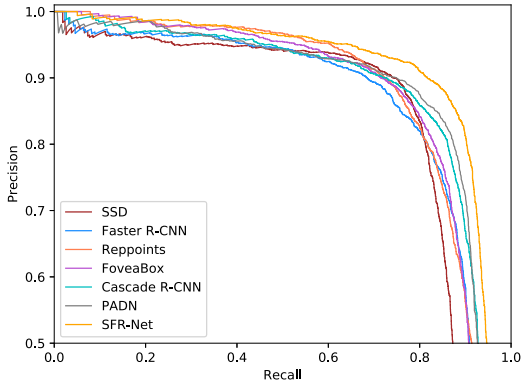


Fig. 17. PR curves of various methods on AIRD.

In order to further verify the detection performance and practical applicability of this method, we conduct experiments on large-scale images. The detection results evaluate the effectiveness of the SFR-Net in the large complex scenes, as shown in Fig. 18. We can observe that the vast majority of aircrafts in

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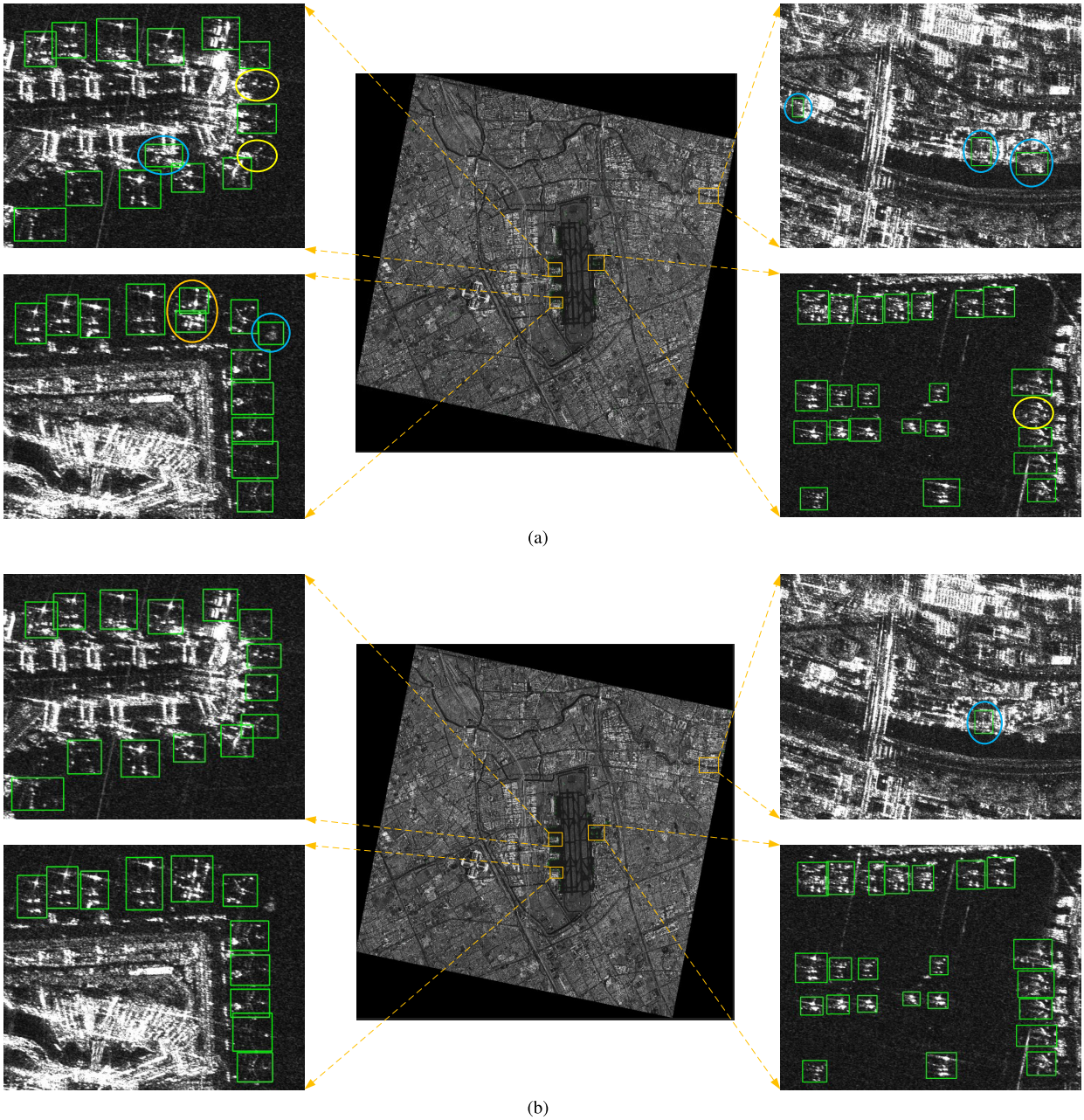


Fig. 18. Detection results of different methods in large-scene SAR images. The green boxes, blue circles, and yellow circles represent the detection objects, false alarms, and missing aircrafts, respectively. (a) Results of PADN. (b) Results of SFR-Net.

the airport are entirely detected, and some components of the aircrafts are rarely separated from the fuselage. Compared with PADN, the proposed algorithm can enhance the salience of the unobvious targets and also reduce the false alarms caused by the buildings with similar scattering at the same time, which exhibits the outstanding advantage in the large complex scenes.

F. Evaluation on Tiered AIRD With Different Bases

The AIRD contains SAR images at different times and multiple azimuths, and the above experiments are based on the multitemporal principle to randomly divide 109 scenes into the training set and the test set. The results in Table V and Fig. 18 indicate that our proposed method plays an effective role in object detection. In order to reflect the generalization ability and robustness of the SFR-Net to different bases, we redivide the dataset according to the bases. The 109 scene images come from 16 bases. We randomly select ten bases and consider all the scene images contained in the ten bases as the training and validation sets, while the scene images in the remaining six bases are regarded as the test set. The same parameters are set, the same way to cut the large image is carried as mentioned in Section IV-A, and the redistributed dataset is called the tiered AIRD.

The comparative experiments with other CNN methods are carried out on tiered AIRD. Table VI indicates that the SFR-Net increases the F1 and $AP_{0.5}$ by more than 4.6% and

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TABLE VI
COMPARISON OF DIFFERENT CNN-BASED METHODS ON TIERED AIRD

Method	Anchor-free	Precision	Recall	F1	$AP_{0.5}$
SSD	×	0.7046	0.5805	0.6366	0.6038
Faster R-CNN	×	0.7237	0.6420	0.6804	0.6704
Reppoints	✓	0.7330	0.6708	0.7005	0.7063
FoveaBox	✓	0.7439	0.6589	0.6988	0.6905
Casacde R-CNN	×	0.7481	0.6347	0.6868	0.6896
PADN	×	0.7425	0.6631	0.7006	0.7089
SFR-Net	✓	**0.7761**	**0.7188**	**0.7464**	**0.7461**

3.7% over other methods, respectively. Simultaneously, the precision and recall have also been significantly improved. Our method leverages the SPRM to extract the scattering characteristics information from the SAR images and develop the connections among scattering points. Besides, the proposed contextual pyramid can highlight the targets with global information and reduce the missing aircrafts. Although the training set and the test set come from completely different bases, SFR-Net still achieves superior performance, which verifies the generalization ability and robustness of the proposed method.

V. CONCLUSION

In this article, a novel anchor-free approach named SFR-Net is designed for aircraft detection in complex scenes. This method leverages the scattering characteristics and the structural information of the aircrafts in SAR images. First, SPRM enhances the connection between strong backscattering points and strengthens the geometric features, which is beneficial to avoid the discreteness between the different components of an aircraft. Second, SFM is introduced to dramatically highlight the meaningful feature and significantly aggregate the semantic information of different layers. Third, CFA can refine the global information characteristics and improve the localization accuracy of the targets by suppressing the background clutter. The satisfactory results of ablation experiments on AIRD demonstrate the superior performance of the proposed modules. In addition, the comparative results with other CNN methods and the outstanding performance in large scenes verify the effectiveness of the SFR-Net.

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